

6. This question is about the uses of gases present in air.

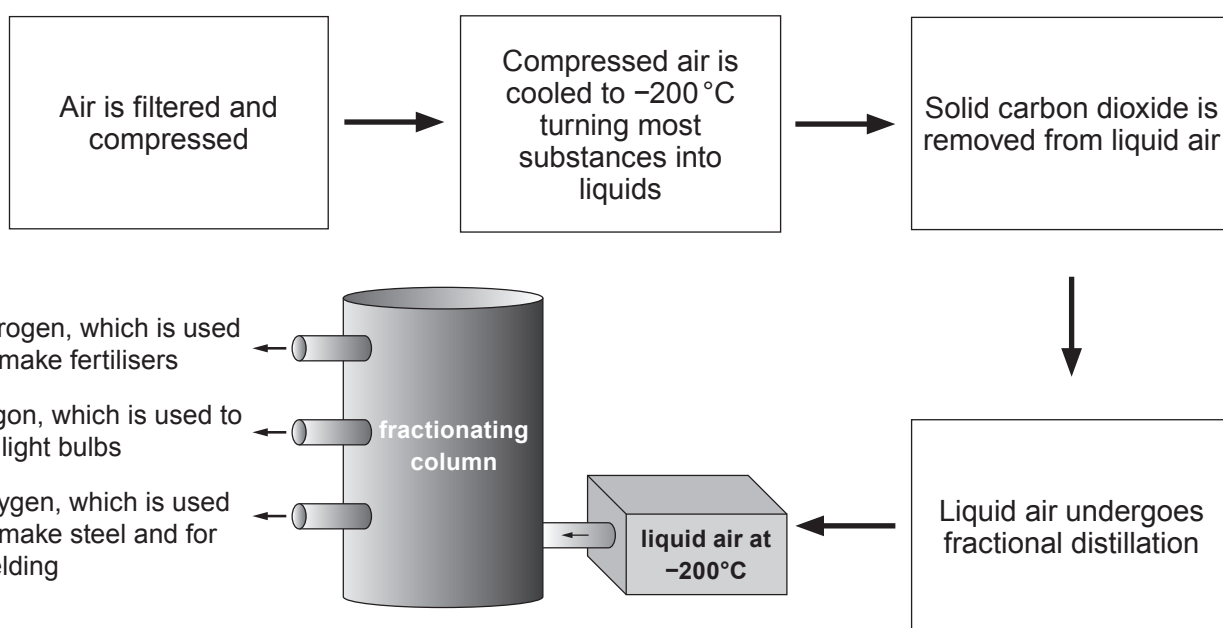
Air is a mixture of different gases. Each gas has different uses. The gases are present in different amounts as shown below.

Gas	Amount present in air	Boiling point (°C)
N ₂	78.04 %	-195.8
O ₂	20.95 %	-183.0
CO ₂	0.03 %	-78.5
Ar	0.93 %	-189.2
Ne	18 ppm	-246.0
He	5 ppm	-268.9
Kr	1 ppm	-152.3
Xe	0.08 ppm	-107.1
H ₂	0.5 ppm	-257.9
CH ₄	2 ppm	-164.0
N ₂ O	0.5 ppm	-88.5

ppm = parts per million

Separating a complex mixture!

Air is firstly compressed and cooled which turns it into a liquid. Carbon dioxide freezes and is removed as solid dry ice. The rest of the mixture goes on to a fractionating column where it is slowly warmed allowing most of the substances to be separated.



- (a) Tick (✓) the **main** reason why hydrogen is not separated during the fractional distillation of liquid air. [1]

hydrogen is a highly reactive gas

☐

only 0.5 ppm of hydrogen is present

☐

hydrogen does not become liquid on cooling to -200°C

☐

hydrogen has a higher boiling point than helium

☐

- (b) Tick (✓) the reason why carbon dioxide becomes a solid during the first part of the process. [1]

carbon dioxide has a boiling point above -200°C

☐

carbon dioxide has a melting point above -200°C

☐

carbon dioxide has a melting point below -200°C

☐

carbon dioxide has a boiling point below -200°C

☐

- (c) Describe, in terms of boiling points, how nitrogen, argon and oxygen are separated during fractional distillation. [3]

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- (d) The noble gases are present in very small quantities in the air. The amount of each gas produced per year is shown in the table.

Noble gas	World production (tonnes per year)
helium	28 000
neon	1 000
argon	700 000
krypton	8
xenon	0.6

Use the data to calculate the number of tonnes of air needed to produce 700 000 tonnes of argon. Give your answer in **standard form**. [2]

Mass of air needed = tonnes

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